Nanopore Compressive Sensing

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December 2025

This document should communicate the current state of the project, it is Work-In-Progress and will change in the future. Is is intended for internal shared use - suggestions, comments or even direct changes are welcomed. Note, that many paragraphs were adopted from the cited work and major rewrite is needed before publishing.

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Abstract

Oxford Nanopore Technologies (ONT) is a company that produces DNA sequencing hardware and software. Their devices produce DNA reads with state of the art length and precision. There is, however, a trade-of between quality and cost - ONT sequencing is orders of magnitude more expensive than short-read sequencing technology used in the clinical setting. Since ONT signal acquisition is costly, we want to examine the theoretical lower bound for the number of measurements required for reliable DNA sequencing.

Model-based compressive sensing is a method for recording a k-model-sparse signal $x \in \mathbb{R}^n$ with (possibly noisy) linear measurements of the form $y = Ax$, where $A \in \mathbb{R}^{m \times n}$ describes the measurement process. Seminal results in model-based compressive sensing show that it is possible to recover the signal x from $m = O(k)$ measurements and that this is tight. Furthermore, there are algorithmic results for signal recovery in nearly linear time[13].

Wavelet transform is a tool for a lossless signal representation with a wavelet dictionary. Since wavelets essentially detect discontinuities, large wavelet coefficients are often propagated down branches of the tree. This suggests that sparse approximations of wavelt-transformed signals have additional structure: they are tree-sparse, by which we mean they are supported on a rooted subtree[5]. Tree-sparsity is an example of model-sparsity.

In this paper we examine and empirically validate the smallest number of measurements required for ONT sequencing signal reconstruction. We achieve that by using a suitable wavelet transform for tree-sparse signal representations and then leveraging model-based compressive sensing framework for signal reconstruction.

1 Introduction

Compressive sensing is a method for recording a signal while taking only a small number of linear measurements. In this setup we are interested in recovering a vector $x \in \mathbb{R}^n$ (the signal) from measurements of form $y = Ax$, where $A \in \mathbb{R}^{m \times n}$ and $y \in \mathbb{R}^m$. Usually, the setup encompasses measurements corrupted by a noise vector e (i.e., $y = Ax + e$), in which case we are interested in recovering a good approximation to x. The main questions in compressive sensing deal with the conditions on A and x that enable efficient, stable recovery from only $m \ll n$ measurements.

Seminal results in compressive sensing show that it is possible to recover a k-sparse signal x (containing at most k non-zeros) from $m = O(k \log n/k)$ linear measurements, as long as the measurement matrix A is chosen to satisfy the *restricted isometry property* (RIP). Moreover, the recovery step can be performed in polynomial time using several algorithms, such as ℓ_1-minimization, Orthogonal Matching Pursuit (OMP), Iterative Hard Thresholding (IHT) or Compressive Sampling Matching Pursuit (CoSaMP). While the bound on the number of measurements m is tight in the noisy k-sparse setting, there are ways to overcome this barrier and improve the "compression rate" even further.

Model-based compressive sensing [4] represents one such approach for reducing the number of measurements. In this framework, we make additional assumptions about the support of the signal x. Instead of considering all d-sparse signals, we limit our attention to a smaller family of k-sparse supports, which we call a *structured sparsity model* $\mathbb{M}_k$. For several sparsity models, it is possible to show measurement bounds of $m = O(k)$.

While model-based compressive sensing succeeds in reducing the number of measurements, it is computationally more expensive. Model-based recovery algorithms rely on the availability of a *model-projection algorithm* - given an arbitrary signal x, a model-projection algorithm returns the best approximation of x in the sparsity model $\mathbb{M}_k$. Recovery is guaranteed only if the projection is exact and the best known model-projection algorithm is $\Omega(nk)$, that means if we treat model-sparsity k as a certain percentage of n, the tree projection is quadratic in n.

However, there is a way to overcome the computational complexity of exact model-projection by extending the sparsity model $\mathbb{M}_k$ to a class of *approximate projections*. This idea led to the new *approximation-tolerant model-based compressive sensing* [12] framework, which improves the performance and achieves nearly linear time.

Our methods For the purposes of discovering the least number of measurements needed for the ONT signal recovery we will use the exact model-projection, which allows us to use simpler recovery algorithms. Although acknowledging, that the same recovery results can be achieved with significantly lower computational time.

This paper relies only on the tree-sparse model for the structured sparsity, since it is empirically shown that this structure is observed in the wavelet-domain representation of natural signal and images.

Background on Oxford Nanopore Technologies signals Long-read genetic sequencing plays a critical role in modern research and clinical setting. ONT devices provide longer reads of DNA and RNA by leveraging their technology for pore and surrounding protein production. One can imagine ONT device takes some DNA sample, allows the DNA to come through the pore and measures the electrical current on that membrane, which can thought of as a 1D signal (just as audio signal). It is common to call the signal produced by ONT devices as *"squiggle"*, referring to its shape. Usual size data produced by sequencing one patient is on the order of magnitude of several TB. Our research is motivated by a practical concern of the total size of sequencing data and by the questioning the redundancy that is encoded in the structure of the data.

2 Results

2.1 Current progress

1. Identification of a wavelet, which shows tree-sparsity structure in wavelet domain of squiggles. Defining suitable tree-sparsity parameter k as a function of n.

2. Employ model-based compressive sensing framework with our wavelet to find measurement and reconstruction routines for successful reconstruction, keeping the number of measurement as low as possible.

Current state of the project I have implemented CoSaMP algorithm for signal reconstruction, with tree-projection done by Condensing Sort and Select Algorithm (CSSA) [3]. The projection algorithm is not exact, which hinders the reconstruction. Also, my current implementation uses Haar wavelet, which isn't the best choice for capturing the tree-structure in the wavelet domain of our signals.

There is however code implementation of an exact tree-projection from [11], which I will integrate into my pipeline.

My next steps

1. Integrate existing exact tree-projection algorithm into my pipeline
2. Implement and benchmark CoSaMP, IHT, OMP with exact tree projection on artificially tree-sparse data. (By first generating a k-tree-sparse projection, which will serve as a original squiggle signal we will measure)
3. Measure the tree-structure of a wavelet-domain coefficients produced from squiggles, compare different orthogonal wavelet families (existing in the 'pywt' Python library). If the structure observed in wavelet-domains of current wavelets is sufficient for significant reduction of measurements (at least 2x), then we can publish the current state. If on the other hand the results aren't satisfactory and we need more than half of measurements for a proper reconstruction, we would need to improve the tree-sparsifying transform (see section about wavelet autoencoder 3.1)

2.2 Expected outcomes

I would like to scope this project as a proof of concept for using model-based compressive sensing with squiggles. As a part of a publication I would like to publish clean codebase. It must not be perfect production ready code, but it should be well structured, readable and follow industry best practices.

Our pipeline Input: squiggle signal z of length n, tree-sparsity k Output: reconstructed signal $\hat{z}$ of length n

The pipeline will

1. Perform the measurement $y = Az$
2. Store as a pair (y, A)
3. Run reconstruction on (y, A) and receive $\hat{z}$
4. Store $\hat{z}$

Proof of Concept validation Once we have $\hat{z}$ we can compare it with the original squiggle z by several metrics. Most straight-forward being the MSE between the two. We can also run basecalling on both signals and compute the Levenstein distance between the resulting DNA sequences. (Andreas would be able to tell us what is acceptable distance from the practical perspective).

Even though our main concern in preserving the same DNA as an output of basecalling, those two metrics are related, so we can use signal MSE as a proxy during pipeline development.

We want to determine the theoretical lower bound for the number of measurements and prove the feasibility of reconstruction. We can claim that our pipeline is working if it can consistently reconstruct the signal returning acceptable DNA sequence from less then 50% of the measurements. The aim is to go under 30%.

3 Discussion

There are a couple of directions for the future improvement of the pipeline.

3.1 Wavelet Autoencoder

Is there an optimal wavelet transform that maximizes the tree-sparsity of squiggles?

There are a couple of results on how to design the custom wavelet based on the signal prior, in order to maximize sparsity. [21] [20] The main idea is, that a wavelet transform is a mechanism that iteratively uses convolution and subsampling operations and so, it is possible to represent a wavelet transform by a convolutional neural network (CNN) and update weights based on a given cost function.

[21] considers the usual wavelets and [20] considers dual-tree wavelet transform.

Dual-tree transform produces more coefficients, which is not a problem for the intermediate representation, additionally it tackles a couple of known issues of classical wavelets, for example - shift-variance, which is absolutely essential if we want to also align the reads and find overlaps.

Related articles: [15] [9]

3.2 Measurement matrix

Class of measurement matrices that show RIP or the model equivalent there of are sub-gaussian. Subgaussian matrices are proven to have (model) RIP with high probability if m (the number of columns) is high enough[4]. Practically speaking we can choose from Gaussian matrix with rows from the normal distribution, Bernoulli matrix with entries 0 and 1, or Rademacher matrix with entries 1 or -1. The matrices must be scaled appropriately, but then those options are supported by thory and are available for us to choose from.

It is known that Fourier subsampling operator also has RIP [10]. Although it isn't apparent to me, that the subsampling operator paired with wavelet transforms would also have RIP, we still need to prove or empirically observe what is possible.

There is also a research on how to train a measurement matrix together with sparsifying transform [8]

3.3 Quantization

Once initial lower bound of full quantization measurements is done and proved to be successful, we can experiment with measuring less bits per measurement.[17][18]

3.4 Speed

For the initial task of proof of concept, the execution speed is not important, since we're trying to find the lower bound for the possible reconstruction, disregarding for now, how fast and complicated the reconstruction is.

However, if the proof of concept is working, we can come back and optimize the performance - use approximate algorithms[12], identify the bottleneck and physical constrains and design a high throughput algorithm, that would leverage available compute. [7]

3.5 Miscelenious

- Tree-based OMP [16]
- Improved Algorithms for model-based recovery [14]
- Tree-approximation and optimal encoding [6]
- Maybe there is a way to use the convex optimization algorithms with structured sparsity [1] [2]
- From denoising to CS [19]

References

[1] Francis Bach et al. *Structured sparsity through convex optimization.* arXiv:1109.2397 [cs]. Apr. 2012. DOI: 10.48550/arXiv.1109.2397.

[2] Sohail Bahmani, Petros T. Boufounos, and Bhiksha Raj. "Learning Model-Based Sparsity via Projected Gradient Descent". In: *IEEE Transactions on Information Theory* 62.4 (Apr. 2016). arXiv:1209.1557 [stat], pp. 2092–2099. DOI: 10.1109/TIT.2016.2515078.

[3] R.G. Baraniuk and D.L. Jones. "A signal-dependent time-frequency representation: fast algorithm for optimal kernel design". In: *IEEE Transactions on Signal Processing* 42.1 (Jan. 1994), pp. 134–146. DOI: 10.1109/78.258128.

[4] Richard G. Baraniuk et al. "Model-Based Compressive Sensing". In: *IEEE Transactions on Information Theory* 56.4 (Apr. 2010), pp. 1982–2001. DOI: 10.1109/TIT.2010.2040894.

[5] Coralia Cartis and Andrew Thompson. "An Exact Tree Projection Algorithm for Wavelets". In: *IEEE Signal Processing Letters* 20.11 (Nov. 2013), pp. 1026–1029. DOI: 10.1109/LSP.2013.2278147.

[6] Albert Cohen et al. "Tree Approximation and Optimal Encoding". In: *Applied and Computational Harmonic Analysis* 11.2 (Sept. 2001), pp. 192–226. DOI: 10.1006/acha.2001.0336.

[7] Tri Dao et al. *FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness*. arXiv:2205.14135 [cs]. June 2022. DOI: 10.48550/arXiv.2205.14135.

[8] Xin Ding, Wei Chen, and Ian J. Wassell. "Joint Sensing Matrix and Sparsifying Dictionary Optimization for Tensor Compressive Sensing". In: *IEEE Transactions on Signal Processing* 65.14 (July 2017). arXiv:1601.07804 [cs], pp. 3632–3646. DOI: 10.1109/TSP.2017.2699639.

[9] Gaëtan Frusque and Olga Fink. "Robust time series denoising with learnable wavelet packet transform". In: *Advanced Engineering Informatics* 62 (Oct. 2024), p. 102669. DOI: 10.1016/j.aei.2024.102669.

[10] Haitham Hassanieh et al. *Nearly Optimal Sparse Fourier Transform*. arXiv:1201.2501 [cs]. Apr. 2012. DOI: 10.48550/arXiv.1201.2501.

[11] Chinmay Hegde, Piotr Indyk, and Ludwig Schmidt. "A fast approximation algorithm for tree-sparse recovery". In: *2014 IEEE International Symposium on Information Theory*. Honolulu, HI, USA: IEEE, June 2014, pp. 1842–1846. DOI: 10.1109/ISIT.2014.6875152.

[12] Chinmay Hegde, Piotr Indyk, and Ludwig Schmidt. "Approximation-Tolerant Model-Based Compressive Sensing". In: *Proceedings of the Twenty-Fifth Annual ACM-SIAM Symposium on Discrete Algorithms*. Society for Industrial and Applied Mathematics, Jan. 2014, pp. 1544–1561. DOI: 10.1137/1.9781611973402.113.

[13] Chinmay Hegde, Piotr Indyk, and Ludwig Schmidt. "Nearly Linear-Time Model-Based Compressive Sensing". In: *Automata, Languages, and Programming*. Ed. by David Hutchison et al. Vol. 8572. Series Title: Lecture Notes in Computer Science. Berlin, Heidelberg: Springer Berlin Heidelberg, 2014, pp. 588–599. DOI: 10.1007/978-3-662-43948-7_49.

[14] Lingxiao Huang et al. *Improved Algorithms For Structured Sparse Recovery*. arXiv:1701.05674 [cs]. Jan. 2017. DOI: 10.48550/arXiv.1701.05674.

[15] Dhruv Jawali, Abhishek Kumar, and Chandra Sekhar Seelamantula. "A Learning Approach for Wavelet Design". In: *ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. Brighton, UK: IEEE, May 2019, pp. 5018–5022. DOI: 10.1109/ICASSP.2019.8682751.

[16] Chinh La and Minh N. Do. "Tree-Based Orthogonal Matching Pursuit Algorithm for Signal Reconstruction". In: *2006 International Conference on Image Processing*. Atlanta, GA: IEEE, Oct. 2006, pp. 1277–1280. DOI: 10.1109/ICIP.2006.312578.

[17] Jason N. Laska and Richard G. Baraniuk. "Regime Change: Bit-Depth versus Measurement-Rate in Compressive Sensing". In: *IEEE Transactions on Signal Processing* 60.7 (July 2012). arXiv:1110.3450 [cs], pp. 3496–3505. DOI: 10.1109/TSP.2012.2194710.

[18] Johannes Benjamin Maly. "Recovery Algorithms for Quantized Compressed Sensing". In: (2019).

[19] Christopher A. Metzler, Arian Maleki, and Richard G. Baraniuk. *From Denoising to Compressed Sensing*. arXiv:1406.4175 [cs]. Apr. 2016. DOI: 10.48550/arXiv.1406.4175.

[20] Daniel Recoskie and Richard Mann. *Gradient-based Filter Design for the Dual-tree Wavelet Transform*. arXiv:1806.01793 [eess]. June 2018. DOI: 10.48550/arXiv.1806.01793.

[21] Daniel Recoskie and Richard Mann. *Learning Sparse Wavelet Representations*. arXiv:1802.02961 [cs]. Feb. 2018. DOI: 10.48550/arXiv.1802.02961.